Valerie Salinas – vjz367
CS 4953
3/20/26

# Annotated Bibliography

[1] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I., 2017. Attention Is All You Need. https://arxiv.org/abs/1706.03762

Introduces the Transformer model that removes recurrence and uses only self-attention mechanism to compute representations of its input and output. The paper breaks down the Transformers model that replaces the common encoder-decoder layers with multi-headed self-attention. Unlike other neural networks like RNNs and CNNs that work the data sequentially word by word Transformers are able to use parallelization that process all the words at once. They use Scaled dot-product attention that computes relationships between words using queries, keys and values. They claim that the transformer will produce faster results and reduce training time making them better than other machine learning models.

[2] Renpu Liu, Ruida Zhou, Cong Shen, and Jing Yang. 2025. On the Learn-to-Optimize Capabilities of Transformers in In-Context Sparse Recovery. arXiv:2410.13981. https://arxiv.org/abs/2410.13981

In this paper they aimed to understand the Transformers in context learning or (ICL) that performs well without the need for further parameter updating. The belief is that Transformers perform pre-conditioned gradient descent which allows them to learn a pre-conditioner during pre-training and then use this in the ICL to expedite and optimize the process. They focus on in-context sparse recovery, where the goal is to reconstruct signals with only a few nonzero elements from limited data. They look at how Transformers implement the L20 algorithm for in-context sparse recovery and show their results.

[3] Rewon Child, Scott Gray, Alec Radford, and Ilya Sutskever. 2019. Generating Long Sequences with Sparse Transformers. arXiv:1904.10509. https://arxiv.org/abs/1904.10509

This paper is focused on ways to change the Transformers that will help generate longer sequences that is not as slow or costly. Because their cost grows quadratically (n^2) with sequence length doubling the sequence is more expensive. They introduce Sparse Attention Transformers, which separate the full attention into shorter parts and are able to handle longer sequences than standard Transformers. To do this they used sparse factorizations of the attention matrix which scale down the cost to $O(n\sqrt{n})$. In addition, they also include updates to the deep learning architecture with adding more layers, recomputing attention only that is needed without saving everything, and use sparse attention kernels. They then show how their work and results of the architecture with model images that use the self-attention that model long length sequences.

Valerie Salinas – vjz367
CS 4953
3/20/26

[4] Piotr Nawrot, Robert Li, Renjie Huang, Sebastian Ruder, Kelly Marchisio, and Edoardo M. Ponti. 2026. The Sparse Frontier: Sparse Attention Trade-offs in Transformer LLMs. arXiv:2504.17768. https://arxiv.org/abs/2504.17768

This paper looks at how well different attention methods perform in large language models (LLMs) and want to know how much we can change or remove to make the models faster without affecting the accuracy. Tested 6 different sparse attention methods, multiple models, and evaluated 9 different tasks. Different methods work best with different tasks, Ex. For reading input use simpler patterns like token-to-global and block-to-block and in generating output use smarter attention to improve performance for more sparsity. They found that Sparse attention enables larger models to outperform smaller models and improve cost and ultimately is important for scaling LLMs.

[6] Dong Liu and Yanxuan Yu. 2025. π-Attention: Periodic Sparse Transformers for Efficient Long-Context Modeling. arXiv:2511.10696. https://arxiv.org/abs/2511.10696

This paper introduces π-Attention, as a new sparse method that is aiming to be better than others such as Ring Attention by handling far away tokens while maintaining efficiency. For π-Attention uses local attention (nearby tokens), periodic skips (jumps to tokens/every k step), and uses adaptive fusion gate (figures out how to combine local and long range info. Their results showed better prediction, fewer FLOPS, and that it scaled well to long sequences. Some issues were that they used a fixed skip patten that could drop performance and that the attention pattern is manually designed making it not at flexible as models that learn. Future plans involve making skip pattern more learnable, combine routing based attention, and apply multimodal models.

[7] Ali Hassani, Steven Walton, Jiachen Li, Shen Li, and Humphrey Shi. 2023. Neighborhood Attention Transformer. arXiv:2204.07143. https://arxiv.org/abs/2204.07143

In this paper they present the Neighborhood Attention Transformer (NAT) which is a sliding window attention mechanism for vision Transformers that focuses on each pixel to its nearest neighbors instead of all pixels globally. They state that it is faster and can be more memory efficient than standard self-attention by reducing the computational complexity from quadratic to linear. The sliding window pattern model can see more areas of an image as layers stack and if a image shifts the features shift as well. They compare NAT to tother sliding window models and show that it performs better, faster and with their NATTEN library easier to parallelize.

[8] Mary Phuong and Marcus Hutter. 2022. Formal Algorithms for Transformers. arXiv:2207.09238. https://arxiv.org/abs/2207.09238

Valerie Salinas – vjz367
CS 4953
3/20/26

This paper focuses on the mathematical computations that make Transformers perform well they present algorithms with explanation and pseudocode. They address the fact that there is not much publication of source and pseudocode within the deep learning community which is the motivation for their paper. They break down tokenization and multiple algorithms used in Transformers with explanation and pseudocode. Then they describe different Transformer architectures and explain how different attention mechanisms work. The paper's purpose is to provide a solid understanding of Transformers and to use the pseudocode as templates for future contribution.

[9] David R. So, Chen Liang, and Quoc V. Le. 2019. The Evolved Transformer. arXiv:1901.11117. https://arxiv.org/abs/1901.11117

In this paper they wanted to know if a machine would produce a better version of the Transformer by searching through many possible architectures automatically instead of manually designing one. The goal was to take Neural Architecture Search (NAS), which was outperforming human designed models, and apply NAS to search for a better Transformer architecture. They introduced Progressive Dynamic Hurdles(PDH), a method that will give more compute to the better models and stop training the weaker models early. They were able to find that it was successful and found a Transformer variation that performed better calling it the Evolved Transformer.

[10] Sainbayar Sukhbaatar, Edouard Grave, Piotr Bojanowski, and Armand Joulin. 2019. Adaptive Attention Span in Transformers. arXiv:1905.07799. https://arxiv.org/abs/1905.07799

Transforms use a fixed attention span which could be costly. This paper proposes using Adaptive Attention Span, a mechanism that allows each attention head to learn its own context length. Instead of using a pre-defined fixed window manually, the model learns how much context it actually needs. This can reduce computation, speed up training, and improve performance. The learned spans reveal that early layers focus locally while later layers capture long range context.

[11] Sainbayar Sukhbaatar, Edouard Grave, Guillaume Lample, Herve Jegou, and Armand Joulin. 2019. *Augmenting Self-attention with Persistent Memory.* arXiv:1907.01470. https://arxiv.org/abs/1907.01470

Since Transformers rely entirely on self-attention over input tokens, they may forget important information because there is no permanent memory. This paper introduces using persistent memory tokens to the Transformer, so instead of only having the input tokens the model also has a set of learnable memory vectors that will act like global storage. This will give the Transformer a memory set to be able to refer to, and it learns what should be stored during training. These memory vectors can help the model remember things giving it a longer range. The results showed that having that additional memory set did make the model better for longer context.

[12] Mikhail S. Burtsev, Yuri Kuratov, Anton Peganov, and Grigory V. Sapunov. 2021. *Memory Transformer*. arXiv:2006.11527. https://arxiv.org/abs/2006.11527

This paper discusses how Transformers struggle with retaining information for long sequences because self-attention only uses information from the current input and doesn't have a structured memory. They propose a Memory Transformer, a modified Transformer that maintains an external memory space which grows over time. They created three extensions MemTransformer, MemCtrl, MemBottleneck and tested on multiple versions. They found that adding memory improves performance, but MemBottleneck performs worse. Ultimately adding memory can improve the Transformers' ability to process global context and complex tasks.

[13] Subhabrata Dutta, Tanya Gautam, Soumen Chakrabarti, and Tanmoy Chakraborty. 2021. *Redesigning the Transformer Architecture with Insights from Multi-particle Dynamical Systems*. arXiv:2109.15142. https://arxiv.org/abs/2109.15142

Transformers work well however they are very large with too many parameters and are slow because attention compares every token with every other token which makes them costly to train and use. The goal for this paper is they want to simplify its attention and feed-forward layers to make it smaller and faster. They introduce a new way to thinking about Transformers and treat it like a multi-particle dynamical system and connect this to the ordinary differential equations (ODEs). They call it the TransEvolve method and instead of tokens attending to each other the tokens move and influence each other like particles in physics. They test this and find that tokens behave like interacting particles reducing computation, parameters and more efficient approximation of attention.

[14] Richard E. Turner. 2026. *An Introduction to Transformers*. arXiv:2304.10557. https://arxiv.org/abs/2304.10557

This paper does an introduction break down of Transformers as a neural network architecture that is used to process sequences or sets of data. Transformers take input tokens and outputs new representations of those tokens which can be used for translation, classification or prediction. The Transformer block structure is broken down into the two main parts: Self-attention (mixes info between tokens), feed-forward network (feed-forward layers refine each token's features). They explain self-attention uses queries, keys, and values to look at each token and to compute how important each token is. They explain and show why transformers are so powerful because they work for so many types of data (text, images, audio), don't require task specific architectures and can process sequences in parallel.

Valerie Salinas – vjz367
CS 4953
3/20/26

[15] Bobby He and Thomas Hofmann. 2024. *Simplifying Transformer Blocks*. arXiv:2311.01906. https://arxiv.org/abs/2311.01906

The paper claims that even though Transformer blocks appear simple they can be sensitive by even the smallest changes when training which could slow it down. They want to know how much from the architecture we can remove before it disrupts performance. They introduce a new version for attention and called it Simplified Attention Sub-block (SAS), which removed some parameters and simplified attention computation. In the Parallel Block Design changed the attention from feed forward to running them in parallel which made the model simpler. They were able to show in their results that the simplified Transformer worked just as well as the standard Transformer and was faster with less resources.

[16] Zichuan Fu, Wentao Song, Yejing Wang, Xian Wu, Yefeng Zheng, Yingying Zhang, Derong Xu, Xuetao Wei, Tong Xu, and Xiangyu Zhao. 2025. *Sliding Window Attention Training for Efficient Large Language Models*. arXiv:2502.18845. https://arxiv.org/abs/2502.18845

The paper addresses the issue with Transformers computation growing quadratically (O(n^2)) with sequence length making them slow, costly and hard to use for long documents. They introduce a new method called sliding window attention training (SWAT), this method uses sliding window attention during training meaning instead of looking at every token each token only looks at nearby tokens. Previously methods used sliding windows only during inference and SWAT applies sliding windowing during training too. They do identify an issue and call it the Attention sink problem which softmax gives high weight to some tokens while others ignored which creates an imbalance in attention. They test SWAT and compare to standard Transformers and find that it has a similar full attention with 50% less memory and 30% faster training.

[17] Manzil Zaheer, Guru Guruganesh, Avinava Dubey, Joshua Ainslie, Chris Alberti, Santiago Ontanon, Philip Pham, Anirudh Ravula, Qifan Wang, Li Yang, and Amr Ahmed. 2021. *Big Bird: Transformers for Longer Sequences*. arXiv:2007.14062. https://arxiv.org/abs/2007.14062

In this paper they developed the BigBird Model, a sparse attention mechanism where instead of every token attending to every other token they limit attention connections which reduces complexity to O(n). BigBird uses three attention patterns the sliding window, global attention and random attention. Using all three approximates full attention efficiently local (captures nearby patterns), global (captures full sequence), random (connects distant tokens). With their results they show that BigBird is a universal approximator and turing is complete. Bigbird can handle sequences 8x longers than standard Transformers and use a lot less memory.

[18] Zhijian Zhuo, Yutao Zeng, Ya Wang, Sijun Zhang, Xiaoqing Li, Jian Yang, Zhou Xun, and Jinwen Ma. 2025. *HybridNorm: Towards Stable and Efficient Transformer Training via Hybrid Normalization*. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*. https://openreview.net/forum?id=NligLHO7yG

The paper states the issue with Transformers and claims its due to how normalization layers are used. They discuss the two common normalization strategies: Pre-Norm being more stable, easier gradient flow but worse final performance and Post-Norm has better performance but instable training which could fail. They propose mixing the two inside a Transformer block and call it HybridNorm. HybridNorm uses QKV Normalization which normalize query, key, value separately and Post-Norm after feed-forward network. They found that HybridNorm showed better results and that using multiple normalization styles made for better accuracy and lower loss.

[19] Brian Lester, Rami Al-Rfou, and Noah Constant. 2021. *The Power of Scale for Parameter-Efficient Prompt Tuning*. arXiv:2104.08691. https://arxiv.org/abs/2104.08691

In this paper they want to know how much task specific information can be learned by tuning only a small number of parameters. They propose prompt tuning as a parameter efficient alternative to full fine tuning. They experiment with different model sizes find that prompt tuning works better as models get larger and almost match full fine tuning and small models benefits less. Comparing to full fine tuning they see that prompt tuning requires <1% of the parameters. Ultimately it is simpler, scalable alternative to fine tuning.

[20] Teodor-George Marchitan, Claudiu Creanga, and Liviu P. Dinu. 2024. *Transformer and Hybrid Deep Learning Based Models for Machine-Generated Text Detection*. arXiv:2405.17964. https://arxiv.org/abs/2405.17964

They begin by stating the need to detect Machine-Generated Text (MGT) now that there have been advances in large language models. They want to take Transformer based architectures and smart hybrid models for detecting MGT. Combining Transformers with CNNs or LSTMs capture local sequential and global features at the same time making them better at detecting MGT. From their experiments they find that Transformers outperform standard models especially RoBERTa based models achieve the best results at finding longer sequences.

[21] Kaleab A. Kinfu and René Vidal. 2025. *Transformers with Joint Tokens and Local-Global Attention for Efficient Human Pose Estimation*. arXiv:2503.00232. https://arxiv.org/abs/2503.00232

In this paper the authors use Transformers to identify where body parts are in an image. They created a model called EViTPose that focuses on important body parts in an image while still maintaining a high accuracy. Another model called UniTransPose focuses on small details and the entire body to understand certain body poses. They combined multiple datasets together so that models would learn better and faster. The results showed that the models were accurate and efficient.

[22] Hemanth Saratchandran and Simon Lucey. 2025. *Enhancing Transformers Through Conditioned Embedded Tokens*. arXiv:2505.12789 https://arxiv.org/abs/2505.12789

There is a problem within Transformers called "ill conditioned attention" that makes training slower and harder. They propose a method called conditioned embedded tokens which changes the input embeddings to make the attention easier to train. By using this method it showed that it stabilizes training and improves performance on image classification, object detection, segmentation and natural language processing. Adjusting the embeddings it can make transformers faster and more reliable without changing the overall architecture. Using this method overall shows it can make transformer train more efficiently.

[23] Ignacio Sastre and Aiala Rosá. 2025. *Memory Tokens: Large Language Models Can Generate Reversible Sentence Embeddings*. In *Proceedings of the First Workshop on Large Language Model Memorization (L2M2)*. Association for Computational Linguistics, 2025. https://arxiv.org/pdf/2506.15001v1

In the paper Large Language models (LLMs) can generate special tokens called memory tokens. These tokens function like reversible sentence embeddings, they can store a sentence in a way and later reconstruct the sentence to its original form. This is important because most embeddings cannot be reversed. Suggesting that this is a new way to store and retrieve text in models. They show in their experiments that this method can help improve LLMs memory and retrieve text accurately.

[24] Eren Unlu. 2024. *Geotokens and Geotransformers*. arXiv:2403.15940. https://arxiv.org/abs/2403.15940

In Geotokens and Geotransformers the authors want to help transformers understand geographic locations better. Transformers use position encodings for work in a sentence but they don't capture actual instances. They created Geotokens, which are tokens that represent locations with latitude and longitude. This is significant because it has the Transformer focus on the tokens based on distance instead of order. With this Transformers can become better when it comes to maps, GPS data or physical location.

[25] Jerome Garnier-Brun, Marc Mézard, Emanuele Moscato, and Luca Saglietti. 2025. *How Transformers Learn Structured Data: Insights from Hierarchical Filtering*. arXiv:2408.15138. https://arxiv.org/abs/2408.15138

Valerie Salinas – vjz367
CS 4953
3/20/26

In this paper they want to understand how Transformers learn and represent structured or hierarchical data. The authors introduce the concept of hierarchical filtering, where early layers capture coarse structure and deeper layers focus on fine details. They showed that Transformers naturally refine hierarchical relationship through layers by analyzing attention weights and by experiments on synthetic and real language tasks. Which would explain why Transformers are good at nested or structed patterns like syntax. In this paper we understand what happens when Transformer modes learn structured data.

[26] Weixin Liang, Lili Yu, Liang Luo, Srinivasan Iyer, Ning Dong, Chunting Zhou, Gargi Ghosh, Mike Lewis, Wen-tau Yih, Luke Zettlemoyer, and Xi Victoria Lin. 2025. *Mixture-of-Transformers: A Sparse and Scalable Architecture for Multi-Modal Foundation Models*. arXiv:2411.04996. https://arxiv.org/abs/2411.04996

Mixture-of-Transformers or MoT is a scalable architecture that uses a sparse mixture of Transformer experts instead of just one large model. In this a router network sends different tokens to only a few relevant experts saving computation and memory. MoT is able to handle large multi-modal datasets while improving performance and accuracy. MoT out performs models of similar size and is able to scale big tasks as well. With MoT we are given a practical way to build an efficient working Multi-modal model.

[27] Joshua Ainslie, Santiago Ontanon, Chris Alberti, Vaclav Cvicek, Zachary Fisher, Philip Pham, Anirudh Ravula, Sumit Sanghai, Qifan Wang, and Li Yang. 2020. *ETC: Encoding Long and Structured Inputs in Transformers*. arXiv:2004.08483. https://arxiv.org/abs/2004.08483

We know that Transformers struggle with long inputs because of self-attention computation time and memory. ETC or Extended Transformer Construction is introduced as a transformer architecture made to handle very long sequences and structured input better than the classic Transformer. ECT uses a global-local attention mechanism to capture overarching context while local tokens attend only to nearby tokens to reduce computation. This lets ETC handle thousands of tokens with much less memory and time than classic transformers. Their experiments show that ETC performs better and more efficiently able to incorporate document structure like sections and paragraphs.

[28] Zhenyu Bai, Pranav Dangi, Huize Li, and Tulika Mitra. 2024. *SWAT: Scalable and Efficient Window Attention-based Transformers Acceleration on FPGAs*. arXiv:2405.17025. https://arxiv.org/abs/2405.17025

In this paper we learn about SWAT, a method and hardware design to accelerate Transformer models with window attention on FPGAs or Field Programmable Gate Arrays. SWAT organizes the data to exploit the local attention pattern which reduces the cost of attention and feed-forward layers compared to classic Transformers. SWAT is also designed to make them run faster and more efficiently on special hardware. SWAT provides a 2x-10x faster interference and lower energy consumption. With SWAT transformers with window attention are more practical.

[29] Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio. 2016. *Neural Machine Translation by Jointly Learning to Align and Translate*. arXiv:1409.0473. https://arxiv.org/abs/1409.0473

In this paper they want to improve neural machine translation so they created soft attention in sequence-to-sequence learning, which at each step of generating a word the decoder focuses on the most relevant parts of the source. Instead of single fixed vector you compute a weighted sum of all source hidden states and this weighted sum becomes a context vector that informs the decoder at each step. With this the model we learn both alignment and translation at the same time without needing explicit alignment labels. This was significant because it laid out the foundation for attention models used in modern deep learning.

[30] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. 2019. *Parameter-Efficient Transfer Learning for NLP*. arXiv:1902.00751. https://arxiv.org/abs/1902.00751

In this paper we learn a new way to fine tune parameters called Parameter-Efficient Transfer Learning that keeps the original model mostly fixed and only trains a small number of extra parameters for each task. This is important because pre-trained models are huge and trying to fine tune all parameters for every model is costly. We can adapt large pre-trained language models to new tasks without updating all their parameters, and instead we can insert small modules called adapter layers between the transformer layers and only train those. Adapter models perform close to full fine tuning while using fewer trainable parameters. With this transfer learning is more memory efficient and preferred when applying the same base model to multiple tasks.